Lesson 1.1: How is Our Ocean Changing?

Location: Classroom

Grade Level: High School

Time: Two 50-minute sessions

Driving Questions

- How will climate change affect what we value about Southern California's ocean?
- How do we critically evaluate information from generative AI?

Lesson Synopsis

In this lesson, students begin exploring how climate change is affecting Southern California's ocean and how different tools — including generative AI models — can help us investigate scientific questions.

Students start by reflecting on their personal experiences with the ocean. They then investigate how climate change is impacting Southern California's ocean by comparing two digital tools: a search engine and a generative AI platform, such as ChatGPT. They are introduced to the CLEAR framework for prompting GenAI. They then work in small groups to use both tools to research ocean change and compare the differences in the information they find.

What students will figure out:

- People and communities value the ocean for its benefits, including mental well-being, economic stability, and biodiversity. Climate change disrupts ocean systems, altering these connections.
- We can use tools like search engines and generative AI to find out more about what is commonly known about how climate change will affect Southern California's ocean.
- To use GenAI effectively, we need to be able to clearly prompt it and stay aware of potential hazards, such as its tendency to hallucinate and make up information.

LEARNING OUTCOMES

By the end of this investigation, students will be able to…	You can assess this using…
1. Explain how generative AI (GenAI) models generate information and compare them to traditional search engines.	Class discussion during the *Investigate* section.
2. Compare and contrast the use of a search engine and a GenAI model, analyzing the differences in the information each tool provides.	Class discussion during the *Navigate* section.
3. Generate questions about what else we need to know to model how climate change will impact Southern California's ocean and its communities.	Student team notes during the *Navigate* section.

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Building towards NGSS and AI Literacy:
HS-ESS2-7: Investigate how gradual atmospheric changes were due to plants and other organisms that captured carbon dioxide and released oxygen. Compare these changes with more recent human-driven changes in carbon dioxide and analyze how these shifts are affecting Southern California's ocean.
HS-ESS2-4: Investigate how human activities have increased carbon dioxide concentrations and thus affect the climate. Explain how these changes are linked to observed and predicted changes in Southern California's ocean.
AI Literacy's Big Idea #3: Learning. Describe how various types of artificial intelligence, such as GenAI, learn by adjusting their internal representations. Compare how these tools differ from traditional search engines.

LESSON OVERVIEW

Section	Time	What Happens
Anchoring Phenomena	15 minutes	Students reflect on their personal experiences with Southern California's ocean and how they believe it is changing due to the impacts of climate change.
Investigate	60 minutes	After brainstorming different methods that they might use to find out more, students are introduced to GenAI and how it works. They are also introduced to the CLEAR framework to help them write effective prompts for GenAI. They then compare a GenAI tool (e.g., Gemini) and a generic search engine (e.g., Google; investigating individual links) to search for more information.
Navigate	15 minutes	After comparing what they found, students reflect on the differences between using a search engine or GenAI to answer questions, identifying the strengths and weaknesses of both.
Navigate	10 minutes	Students reflect on what else they might need to know in order to develop a model that shows how climate change is affecting Southern California's ocean.

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MATERIALS

- Lesson 1.1 Slides
- Science Journal
- A screen to view the slideshow with audio (projector, Promethean, etc.)

SUPPLIES

- (1) Laptop per group
- (1) Science journal per student

BEFORE YOU START TEACHING

1. Review CLEAR Framework to support students in prompting GenAI (see Slide 14-15; Lo, 2023; Teacher Facilitation Guide)**.**
2. Throughout the program, students will be working in teams of three or four. It may be helpful to ensure that each team has at least one team member who is comfortable with new technological tools, so we recommend selecting the student teams before the start of module 1.1.
3. Make sure that you have a way to project a video with sound
4. The science journal is intended to be a digital notebook. However, if you decide to print copies for students, they may need a writing instrument (pen, pencil, etc.).

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: This lesson builds a foundation for the rest of the module and invites students to investigate and collect information for how Southern California's ocean is changing. As the first lesson in the module, the goal is not to establish ideas related to the disciplinary core ideas (DCIs), but to elicit student ideas and experiences. Some DCIs students will be thinking about include:

- **HS-ESS2-7**: Gradual atmospheric changes were due to plants and other organisms that captured carbon dioxide and released oxygen.
- **HS-ESS2-4**: Changes in the atmosphere due to human activity have increased carbon dioxide concentrations and thus affect the climate.

Students will encounter the terms: socio-ecological systems, generative artificial intelligence, large language models, search engines. They will encounter several terms in previous grade bands in NGSS performance expectations. These terms include climate change, global warming, weather, carbon dioxide, and carbon cycle. Students do not need to have a complete understanding of these ideas, as they will develop understanding across the unit.

Where We Are NOT Going: Students will engage with their communities outside of the classroom to bring some relevant experiences to the classroom that may support the class to figure out the phenomenon. We will not fully investigate the impacts of climate change from around the world or dive deep into investigations on particular climate change impacts, such as coastal erosion, ocean acidification, or global warming.

LEARNING SEQUENCE

Lesson plan notes are also on the slide notes.

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***ANCHORING PHENOMENON*: Introducing Today's Session (15 minutes)**

1. Open the slideshow to **Slide 1** and welcome students into the classroom and inform them that over the next few weeks, they will help pilot a new curriculum.

2. Advance to **Slide 2**, where students will be introduced to **Agents for Inclusive Science Communication** (AISciComm), the program that they will help pilot.

3. Afterwards, move on to **Slide 3**, and showcase who is involved in creating the program.
    - Inside the Outdoors
    - Orange County Department of Education
    - University of California Irvine
    - University of North Carolina at Chapel Hill

    The focus of this program will be on how climate change impacts Southern California's ocean and the values that local communities place on the ocean system. Students will explore systems and systems modeling, gathering information through community interviews as well as GenAI and search engine research.

4. Once students have been introduced to the program, advance to **Slide 4** and share an overview of module 1.1.

5. On **Slide 5,** ask students to get into groups of three (preferably). These will be the groups that they work with through the remainder of their project.

    If you prefer, pre-select which students will be working together.

6. Advance to **Slide 6**, where students are introduced to their digital Science Journal. Their Science Journal will be used as a central document to share instructional tasks, capture their reflections, and provide brainstorming documents throughout the lessons.

7. Advance to **Slide 7** and tell students that their first team task is to reflect on some initial questions. Invite students to reflect in their groups on the following questions:

    - How do you interact with the ocean? What values do you place on the ocean?
    - How does your community interact with the ocean? What values do they place on the ocean?

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- What impact do you think climate change will have on the way communities interact with the ocean?

8. As students discuss in their groups, walk around and observe the topics being discussed. If needed, provide additional probing questions to keep the conversation going.
    - Do you think the oceans are important to you or the people you know? Why or why not?
    - What activities might you or the people you know do at the ocean?
    - Could the activities people do change over time with climate change? How?
    - What have you heard happen to our ocean communities due to climate change events (e.g. rising sea levels, extreme weather)?

9. As time winds down, invite students to gather back together for a class discussion. Advance to **Slide 8** and share the following questions.
    - What did you and your team discuss?
    - How is the ocean important to you or others in your community?
    - What types of challenges may the ocean face?

10. Advance to **Slide 9** to give students an overview of the day: Researching how Southern California's ocean may be impacted by climate change through Gemini (a GenAI tool) and Google (a search engine).

INVESTIGATE: **An Introduction to Generative AI; Using Generative AI for Information Search (75 minutes)**

1. Explain to students that in order to investigate how climate change is impacting the ocean, they will be exploring two types of tools: (1) a search engine like Google, and (2) a GenAI tool like Gemini.

2. Since they are already familiar with searching for information through platforms like Google, tell students that they will begin by learning more about generative artificial intelligence (or GenAI).

3. To gauge what students may know about GenAI, advance to **Slide 10** and ask:
    - Do you know any platforms that are considered Generative AI? What are they?
    - What do these platforms do? What can they generate?

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Students may share examples like ChatGPT, Bing, MagicSchoolAI, etc., as potential platforms and share that they can provide text, images, or responses to queries.

4. Advance to **Slide 11** and share that GenAI is a type of machine intelligence that can create (or generate) new content, such as text, images, music, or video.

5. Advance to **Slide 12** and inform students that they will interact specifically with Large Language Models (LLM), a form of GenAI that can generate text. To understand how LLMs work, they will explore it by watching a video. [0:00 - 1:12; watch the video from the beginning to the **1:12** mark]

6. After students have watched the video, move on to **Slide 13** and ask the following questions:
   - What are large language models? What makes them "large"?
     Large Language Models are trained on a very large amount of text data, often pulled from the Internet. If a human were to read the text used to train the current version of ChatGPT, it'd take about 3500 years.
   - If we were building a chatbot model to sound like you, what types of data could the chatbot be trained on?
     The chatbots would be trained on your daily conversations and written text, so they can predict the words to sound like you.

7. Explain to students that their team's task is to search for information on how climate change is impacting the ocean in Southern California using a large language model (e.g., Gemini) and Google search.

8. Before they begin searching, advance to **Slide 14**, where students will see the CLEAR framework, a framework used to effectively create question prompts for GenAI. Share that an effective question follows the CLEAR framework and is:

   - **C**oncise: uses only essential words
   - **L**ogical: has a logical and structured order to your prompt
   - **E**xplicit: provides exact instructions about your desired result
   - **A**daptive: provides room to further the conversation or make improvements to your question
   - **R**eflective: provides space to interpret the result and make sense of it. Does the provided answer make sense?

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- **Review Teacher Facilitation Guide (Under Content Review and Core Concepts) for guiding questions on CLEAR Framework.**

9. To see an example of a question that uses the CLEAR framework, advance to **Slide 15**.

10. Advance to **Slide 16** and share a final few tips that students should keep in mind while searching for information:

    - Consider the accuracy of the source. Is the information provided accurate or outdated?
    - Check additional sources and make sure that the sources are reputable.
    - If the response is unclear, ask for clarification and examples.
    - Test varying prompts or search terms to make sure that the answers are consistent.

11. Advance to **Slide 17** and invite students to start using both GenAI (Gemini) and Google to investigate how climate change is impacting the ocean.
    Remind students to keep the following questions in mind during their search:
    - How has climate change impacted the ocean in Southern California over time?
    - How might climate change impact the way communities connect with the ocean?

12. If student teams have not done so, invite them to test both platforms using similar questions to investigate the similarities and differences in responses.

NAVIGATE: **Reflecting on using GenAI for information search (15 minutes)**

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1. As students finish their information search, bring the class back together to share what they have learned.

2. Move forward to **Slide 18** and invite students to share their experiences on the differences in the interactions and outputs between using Gemini versus Google.

   Consider the following questions to facilitate student discussion:
   - What platforms did you use?
   - Was the way you searched for information different depending on the platform? How?
   - What type of responses did you get?
   - How did you evaluate if the information provided was accurate and reliable?
   - What do you think are the strengths and weaknesses of a GenAI platform like Gemini and a search engine like Google?

NAVIGATE: **Reflecting on what we know (10 minutes)**

1. Advance to **Slide 19** and invite students to add to their science journal, reflecting on what else they might need to know in order to develop a model that shows how climate change is affecting Southern California's ocean.

   As you finish up for the day, thank the students for their time and ask them to reflect on these questions on **Module 1** of their **Science Journal**:

   - How has your understanding of how climate change affects Southern California's ocean changed after completing Module 1.1?
   - What information are you missing that can help you construct a model of how climate change is impacting Southern California's ocean?
   - What other sources or people could help you learn more information about what you want to know?

2. Advance to **Slide 20**. Thank the students for their time today. Recap with students that today, they used GenAI and search engines to gather information. Their goal by the end of their time with this project will be to construct a model showing how climate change is impacting Southern California's ocean. To do so, their next step is to gather additional information by using a different source of data collection, such as interviews.

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